Wöden Kusner

Curriculum Vitae, October 2026

Citizenship USA

Research Interests Applied Geometry and Optimization, Topology of Configuration Spaces, Topological Data Analysis, Approximation Theory, Representation Theory, Statistical Mechanics and Condensed Matter Physics.

Education

University of Pittsburgh, Pittsburgh, PA USA
Ph.D., Mathematics — August 2014
Advisor: Professor Thomas C. Hales
Dissertation: *Bounds on packing density via slicing*
M.A., Mathematics — December 2010

Haverford College, Haverford, PA USA
B.S., Mathematics — May 2007
Advisor: Professor John J. Flynn
Thesis: *Results in sphere packing density*

Certificates

University of Georgia, Athens, GA USA
Certificate in Diversity and Inclusion — May 2021

Vanderbilt University, Nashville, TN USA
STEM Teaching Specialization — June 2020
Certificate in College Teaching — May 2019

Professional Development

- CEL Cohort, Lawrence, 2026–2027
- REFLECT, Lawrence CTE, Winter 2024

Academic Appointments

Adjunct Assistant Professor of Mathematics — 2025–
Director, Quantitative Resource Center
Department of Mathematics
Lawrence University

Instructor of Mathematics — 2023–2025
Department of Mathematics
Lawrence University

Assistant Professor (NTT) — 2020–2022
Department of Mathematics
University of Georgia

Assistant Professor (NTT) — 2017–2020
Department of Mathematics
Data Science Institute Faculty Affiliate
Vanderbilt University

Postdoctoral Associate — 2017–2020
Center for Constructive Approximation
Vanderbilt University

FWF Postdoctoral Researcher 2014–2017
Institute of Analysis and Number Theory
Graz University of Technology

Visiting Scholar
Isaac Newton Institute (Aberystwyth) 2023
Fields Institute 2021
ICERM, Brown University 2015, 2018
Erwin Schrödinger International Institute, University of Vienna 2014

Publications

- Critical radii of kissing configurations: algebraicity, balance, and twelve spheres above $r = 1$. (*in preparation*).
- with G. Buck, R. Kusner. Stopper knots. (*in preparation*).
- with G. Dietler, E. Rawdon, R. Kusner, P. Szymczak. Chirality for crooked curves. (*in revision*). arXiv:2004.10338
- with R. Kusner. A Gordian pair of links. *Geometriae Dedicata*, 2023. arXiv:1908.05610
- with T. Hales. Packings of regular pentagons in the plane. (*submitted*). arXiv:1602.07220
- with J. Brauchart, P. Grabner, J. Ziefle. Hyperuniform point sets on the sphere: probabilistic aspects. *Monatshefte für Mathematik*, 2020. arXiv:1809.02645
- with J. Brauchart, P. Grabner. Hyperuniform point sets on the sphere: deterministic aspects. *Constr Approx*, 2019. arXiv:1709.02613
- with R. Kusner, J. Lagarias, S. Shlosman. Configuration spaces of equal spheres touching a given sphere: the twelve spheres problem. *Bolyai Society Mathematical Studies: New Trends in Intuitive Geometry*, 2018. arXiv:1611.10297
- with Y. Kallus. The local optimality of the double lattice packing. *Discrete Comput Geom*, 2016. arXiv:1509.02241
- On the densest packing of polycylinders in any dimension. *Discrete Comput Geom*, 2016. arXiv:1405.0497
- An upper bound on packing density for circular cylinders of high aspect ratio. *Discrete Comput Geom*, 2014. arXiv:1309.6996

Talks and Conferences

- CTE Lunch and Learn, Lawrence University: *A discussion about the new Quantitative Resource Center and our perspectives of math.* 10/28/25
- Workshop on Institutional Approaches to Quantitative Skill Support, Carleton College. 4/4/25
- Society of Physics Students Journal Club, Lawrence University: *Chirality for crooked curves.* 1/25/24
- AMS Fall Southeastern Sectional, Special Session on Discrete Geometry and Geometric Optimization, South Alabama: *Critical kissing spheres.* 10/15/23
- INI GeomPack satellite programme / 5th International Conference on Packing Problems, Aberystwyth: *Building Gordian links.* 8/22/23
- ESI Workshop: Optimal Point Configurations on Manifolds (hybrid). 1/22
- GeoFest 2020 (K. Bezdek 65th), Veszprém: invited; cancelled (COVID). */20
- Carolina Geometry Seminar: *Measuring chirality with the wind.* 4/8/21
- Points Seminar: *Measuring chirality with the wind.* 4/7/21
- Auburn University Colloquium: *Chirality and hydrodynamics.* 11/13/20
- UGA Geometry Seminar: *Building Gordian unlinks.* 11/6/20
- UGA Topology Seminar: *Chirality and hydrodynamics (à la Lord Kelvin).* 8/31/20

- BIRS–CMO Workshop 19w5142, Soft Packings, Nested Clusters, and Condensed Matter, Oaxaca: *Gordian unlinks.* 10/3/19
- 4th International Conference on Packing Problems, Yale. 6/*/19
- Topology and its Applications, WKU: *Gordian configurations (II).* 7/17/18
- ICERM, Brown: *Gordian configurations (I).* 4/11/18
- ICERM, Brown: *Computing discrepancy.* 3/9/18
- Aspen Center for Physics. 6/*/17
- Montanuniversität Leoben: *Critical packings & the radius function.* 6/2/17
- CEIM, Universidad de Cantabria: *Critical packings (in the sphere).* 4/22/17
- JMM, Discrete Geometry & Convexity: *Critical packings, rigidity, & the radius function.* 1/6/17
- TU Graz, Fall School: *Critical packings, rigidity, & the radius function.* 9/30/16
- AIM Workshop: *Configurations of spheres.* 9/22/16
- ICERM Workshop, Brown. 9/*/16
- ACG Seminar, Pittsburgh: *Configurations of spheres.* 8/25/16
- MCQMC, Stanford: *Configurations of points w.r.t. discrepancy & uniform distribution.* 8/17/16
- MSRI: *Critical packings, rigidity, & the radius function.* 8/4/16
- MSRI Summer Research. */16
- Institut Henri Poincaré Workshop. 6/*/16
- AMS Spring Southeastern Sectional, Special Session on New Developments in Discrete and Intuitive Geometry: *Configurations of points w.r.t. discrepancy & uniform distribution.* 3/6/16
- Advanced Topics Seminar, TU Graz: *Configurations of spheres.* 1/22/16
- Zahlentheoretisches Kolloquium, TU Graz: *Problems with packing periodicity.* 12/11/15
- ICERM, Brown: *Can rods pack space more densely than disks the plane?* 4/28/15
- ICERM, Brown: *Spherical discrepancy.* 4/9/15
- TU Graz: *Computing spherical cap discrepancy: proof of concept.* 1/22/15
- Guest Lecture, TU Graz: *Introduction to packing problems.* 1/19/15
- Large Structures Seminar, Aalto: *Packing density bounds in higher dimensions.* 11/22/14
- ESI Workshop: *A brief analysis of regular pentagon packings in the plane.* 8/27/14
- IAS–PCMI Researcher: Mathematics and Materials. 6/*/14
- Oberwolfach: *Packing polycylinders.* 6/*/14
- Dissertation Defense, Pittsburgh: *Bounds on packing density via slicing.* 5/22/14
- Seminar, TU Graz: *Packing density bounds via slicing.* 5/8/14
- Erdős Memorial Lectures, Memphis: *Polycylinder density in higher dimensions.* 3/*/14
- Fields Institute: Workshop in Discrete Geometry. 11/*/13
- GSS, Pittsburgh: *Some packing problems & an upper bound.* 3/28/13
- A&S Graduate Expo, Pittsburgh: *Packing cylinders with high aspect ratio.* 3/23/13
- ACG Seminar, Pittsburgh: *An upper bound on packing density for circular cylinders with high aspect ratio.* 2/12/13
- Topological Dynamics Workshop, Newton Institute: *Packing circular cylinders.* 11/*/12
- IMA Summer School in Topological Methods, Penn. 7/*/11
- Graduate Algebra, Combinatorics and Geometry Seminar, Pittsburgh: *The Jones polynomial and the Kauffman bracket*; *Category theory* II–V (products and limits; slice and comma categories; limits informally and formally; representable functors). 2010–2014
- Senior Thesis Defense, Haverford: *Results in sphere packing density.* 5/*/07

Other Works

- *Of Instruments*: working interactive mathematical instruments (slide rules, Napier's bones, planimeters, verniers, and others), 2026
- Untitled work in *Frayed Knot* at The Museum of Everyday Life, 2020
- Cover art for *New Trends in Intuitive Geometry*, 2018

Honors & Awards

- UGA Teaching Academy Fellow, 2021–2022
- Member of the Instructional Corps for the UGA Department of Mathematics, which received the 2022 Regents' Teaching Excellence Award from the University System of Georgia
  State-wide award to a department or program for excellence in teaching and in service to students.
- Work featured in Die Presse: Science and Innovation, 2017
- University of Pittsburgh Honors Convocation, 2013, 2014
- Outstanding Lecture/Presentation, University of Pittsburgh Graduate Expo, 2013
  Award for the best lecture by a graduate student in the Natural Sciences.
- K. Leroy Irvis Fellow, University of Pittsburgh, 2009, 2012
  These highly competitive fellowships are designed to meet the critical need to recruit underrepresented minority graduate students to the University of Pittsburgh, to support and retain them, and eventually enhance their presence in the professorate.
- Bronze Presidential Service Award for AmeriCorps Volunteer Service, 2008
  The President's Volunteer Service Award is a civil award bestowed by the President of the United States to honor volunteers who give hundreds of hours per year.

Teaching

Lawrence University, Appleton, WI USA

Adjunct Assistant Professor	9/25–
MATH 103: Preparation for Calculus	Fall 26
MATH 140: Calculus	Spring 26
MATH 102: Foundations in Math *TA: Ellison McLane*	Winter 26
MATH 103: Preparation for Calculus	Fall 25
MATH 591: Random Walks & Electrical Networks	Fall 25
Instructor	9/23–8/25
MATH 103: Preparation for Calculus	Spring 25
MATH 103: Preparation for Calculus *TA: Eli Hooker-Reese*	Winter 25
MATH 591: Differential Topology	Winter 25
MATH 103: Preparation for Calculus *TA: Elijah Pile*	Fall 24
MATH 103: Preparation for Calculus	Spring 24
MATH 102: Foundations in Math *TA: Hasif Ahmed*	Winter 24
MATH 103: Preparation for Calculus *TA: Mina Mandegar*	Fall 23

University of Georgia, Athens, GA USA

Assistant Professor (NTT)	8/20–12/21
Instructor for MATH 2250: Calculus I (2 sections)	Fall 21
Instructor for MATH 2260: Calculus II	Spring 21
Instructor for MATH 2250: Calculus I (2 sections)	Fall 20

Vanderbilt University, Nashville, TN USA

Assistant Professor (NTT) 8/17–8/20

- MATH 2300: Multivariable Calculus — Spring 20
  grader: Emma Francis
- MATH 2300: Multivariable Calculus (2 sections) — Fall 19
  grader: Emma Francis
- Undergraduate Summer Research (3 students) — Summer 19
- MATH 3641/5641: Mathematical Statistics — Spring 19
  grader: Xiaolu Qian
- MATH 2300: Multivariable Calculus — Fall 18
  grader: Xiaolu Qian
- MATH 1010: Probability & Statistical Inference I — Fall 18
  TAs: Jamison Donohue, Min Soo Kim, Shaoyang Zhou; grader: Jessica Wang
- MATH 3641/5641: Mathematical Statistics — Spring 18
- MATH 1011: Probability & Statistical Inference II — Spring 18
  TAs: Junying Fang, Xiayimei Han, Qixuan Wang
- MATH 1010: Probability & Statistical Inference I — Fall 17
  TAs: Amber Teetsel, Qixuan Wang, Helen Zhou

Graz University of Technology, Graz, AT

Lehrbeauftragter 10/14–1/15, 3/16–6/16

- Instructor for MAT.670: Packings, Lattices and Configurations — Summer 16
- Assistant for MAT.902: Höhere Analysis — Winter 14

University of Pittsburgh, Pittsburgh, PA USA

Teaching Fellow 9/10–12/11

- Math 0220: Calculus I (2 sections) *E. Arrington, M. Chen* — Fall 11
- Math 2700: Graduate Topology *P. Gartside* — Fall 11
- Math 1700: Topology *B. Ion* — Spring 11
- Math 1410: Foundations of Mathematics *T. C. Hales* — Spring 11
- Math 1250: Abstract Algebra II *J. Wheeler* — Spring 11
- Math 0230: Calculus II *D. Vassilev* — Fall 10
- Math 0220: Calculus I (2 sections) *B. Michael, E. Trofimov* — Fall 10

Teaching Assistant 9/09–8/10

- Instructor for Math 0120: Business Calculus — Summer 10
- Math 0120: Business Calculus (3 sections) *C. Duane-Lennard, B. Michael* — Spring 10
- Math 0240: Calculus III (3 sections) *J. Wheeler, J. Arciero* — Fall 09

MENTORING

- Research Mentor (LU Research Fellows), Ellison McLane: discrepancy theory, Summer 2025 (Lawrence)
- Organizer and mentor, Grad School Boot Camp (Math GRE practice exams and application preparation) for Mina Mandegar, Ellison McLane, and Binh Vu, Summer 2025 (Lawrence)
- Directed studies (MATH 591): Differential Topology, Winter 2025; Random Walks and Electrical Networks, Fall 2025 (Lawrence)
- Letters of recommendation for more than 20 students (2018–2025): graduate programs in mathematics, statistics and financial mathematics; summer research and internships (e.g. Caltech SURF, DOE SULI, NREIP); medical school; transfer, study-abroad and off-campus programs; employment, including an FBI background-investigation interview (2020)

- Research Mentor for Kevin Hu, B.S. Highest Honors 2020 (Vanderbilt)
- Honors Thesis Committee for David K. Zhang, B.S., Founders Medal 2019 (Vanderbilt)
- Dissertation Committee for Oleksandr Vlasiuk, Ph.D. 2018 (Vanderbilt)
- Research Mentor for Jonas Ziefle (Graz)

Service

University-focused

- Director: Quantitative Resource Center (Lawrence), 2025–
  Peer tutoring and mentoring of peer tutors, the Quantitative Learning Assistant (QLA) program, the QRC Resource Directory, quantitative-skills coordination across departments, individualized support for case-managed students, and open office hours for students, faculty, and staff.
- Member: Quantitative Reasoning GER working group (learning outcomes), Lawrence, 2025

Teaching-focused

- Co-organizer: UGA Math Learning and Teaching Seminar, 2022
- UGA Calculus Exam Committee, 2021
- LS-PAC MODELS Mentor, 2021–2023
  The LS-PAC MODELS mentorship program is a nationwide opportunity; anyone belonging to a historically underrepresented group in STEM at an institution with an LSAMP program can participate.

Research-focused

- Active referee and reviewer for journals and scientific bodies, including BIRS, ESF, NSF, NWO; Ann. of Math., Constr. Approx., DCG, Exp. Math., GCOM, IMRN, JKTR, Proc. Roy. Soc. Edinburgh A, SIAM J. Discrete Math., WADS
- Reviewer: NSF Graduate Research Fellowship Program panel (Mathematical Sciences), 2022
- Co-organizer: AMS Special Session on Discrete Geometry and Geometric Optimization, Fall Southeastern Sectional, 2023 (postponed from 2021)
- Co-organizer: Shanks Workshop (Vanderbilt), 2019
  Proposed, organized, coordinated, and assisted in funding ($5,000–$10,000) the Shanks Workshop on Computations and Linear Programming Bounds for Energy, Packing and Covering.
- Co-organizer: Fundamental Lemma → Discrete Geometry → Formal Verification, 2018
  Proposed, organized, coordinated, and acquired funding ($25,000–$50,000).
- Organizer: Computational Analysis Seminar (Vanderbilt), 2017–2020
  Managed the weekly seminar run out of the Center for Constructive Approximation.
- Website Maintainer: CCA and CA Seminar (Vanderbilt), 2017–2020
  Maintained and updated the website for the Center for Constructive Approximation.
- Organizer: Graduate Seminar in Algebra, Combinatorics and Geometry (Pittsburgh), 2010–2014

Community-focused

- Historic Central Neighborhood Association (Appleton): Secretary, 2021–; Core Team, 2024–
- Edison/Columbus Safe Routes to School Transition Subcommittee (Appleton Area School District), 2025

- Neighborhood Advisory Team, Zion Lutheran Affordable Housing Project (Appleton), 2024
- Neighborhood volunteer: Columbus community garden (2025); Appleton Porchfest (2026)
- Fox Valley Data Exchange, data lakehouse specification working group, 2025
- Community Data Hub Project / IFC Data Team (Fox Cities), 2021–2022
  Imagine Fox Cities is an inclusive community-wide initiative created to be intentional (i.e. data driven) about shaping the future of the Fox Cities when it comes to well-being.
- Kettering Foundation Research Exchange, 2021
  The Kettering Foundation is a nonprofit operating foundation rooted in the American tradition of cooperative research. Kettering's primary research question is, what does it take to make democracy work as it should?
- Co-organizer and panelist: BiPOC Narratives in Medicine (Haverford), 2021; Alumni Weekend panelist (Haverford), 2021
- Leadership Committee: Multicultural Alumni Action Group (Haverford), 2020–2024
  The Multicultural Alumni Action Group exists to ensure an inclusive community where all students and alumni can thrive.

Graduate student service (Pittsburgh)
- Representative, Dietrich School of Arts and Sciences Council, 2012–2014
- Delegate, Arts and Sciences Graduate Student Organization, 2011–2014
- President, Mathematics Graduate Student Organization, 2013–2014
- Treasurer, Mathematics Graduate Student Organization, 2011–2013
- Treasurer, SIAM University of Pittsburgh Chapter, 2010–2011

Memberships AMS, MAA, APS

Technical Skills TEX (LATEX, BIBTEX), Mathematica, Python, Julia, HTML/CSS/JavaScript, Git/GitHub; AI-assisted research and development (Claude, Claude Code).

Languages English (native), German (intermediate), Spanish (elementary).

References Please contact me for references.